DeepScientist Citation Audit Agent

Evidence PDF: QQ interaction, backend logs, detailed test outputs, and boundary cases

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Required evidence	Location in this PDF
QQ / WeChat real interaction screenshots	Page 2: command, progress heartbeat, DONE summary, and END OF RUN screenshots
Agent backend running screenshots	Page 3: tool_calls.jsonl and citation_verification.csv screenshots; Page 11 lists extra screenshot targets
At least 5 test samples with inputs and outputs	Pages 4-6: five real PDF cases, normalized inputs, output directories, artifact counts, and representative output rows
At least 2 failure or boundary cases	Pages 9-10: FinRL all-Yellow boundary case and GW150914 Red metadata-conflict case

33 CSV files	**201** Checked claims	**1408** Tool calls
985 Evidence items	**24** Multi-review reports	**24** Case reports

Green / Yellow / Red summary: **61** / **138** / **2**. Scope: deepscientist_15x_campaigns, deepscientist_20x_campaigns, and outputs/runs. The /related module is excluded.

Yellow is intentionally frequent because the verifier is conservative: an existing paper is not marked Green unless concrete support is found.

1. QQ Real Interaction Screenshots

The screenshot below comes from a complete QQ demo workflow: the user sends an /official command, the bot reports status checks, progress heartbeats, citation audit, multi-reviewer, memory update, DONE summary, and END OF RUN.

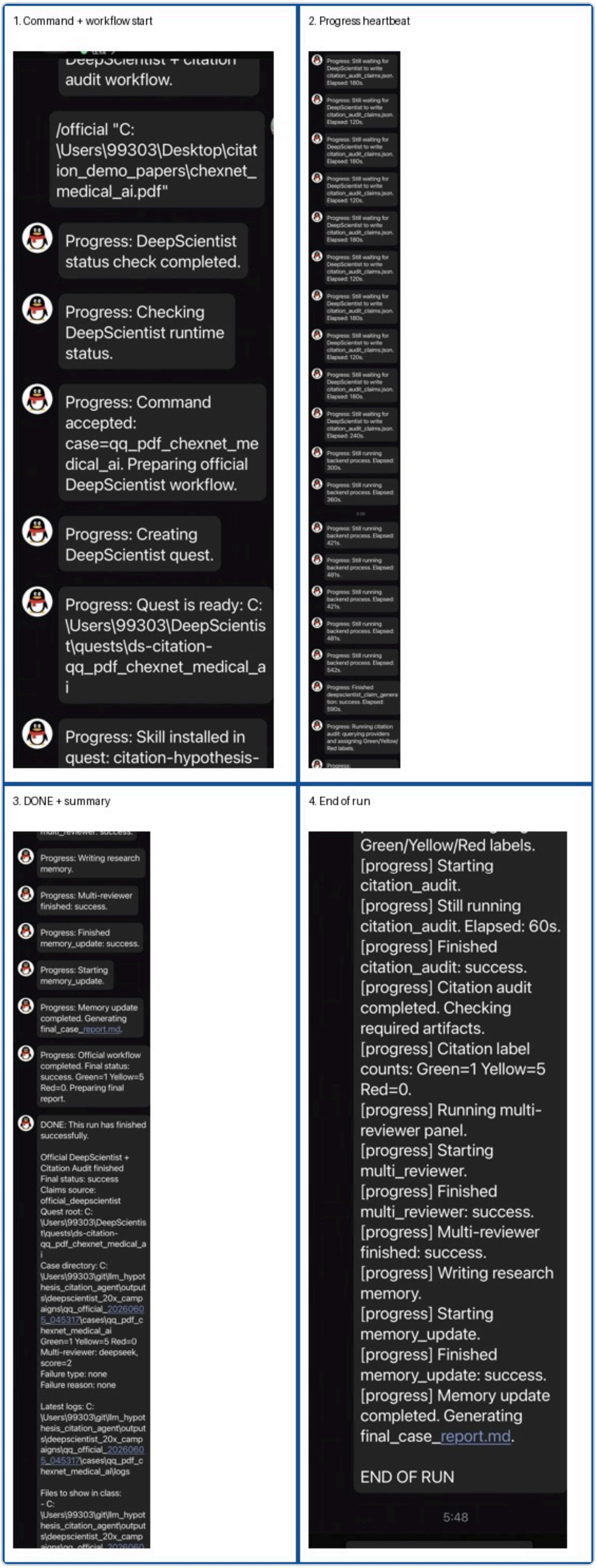

Figure 1. QQ evidence overview: command accepted, progress heartbeat, DONE summary, and END OF RUN.

2. Agent Backend Running Screenshots

These screenshots show that the system is not a prompt-only chat. It writes auditable tool logs, citation tables, evidence items, and evidence chains for every run.

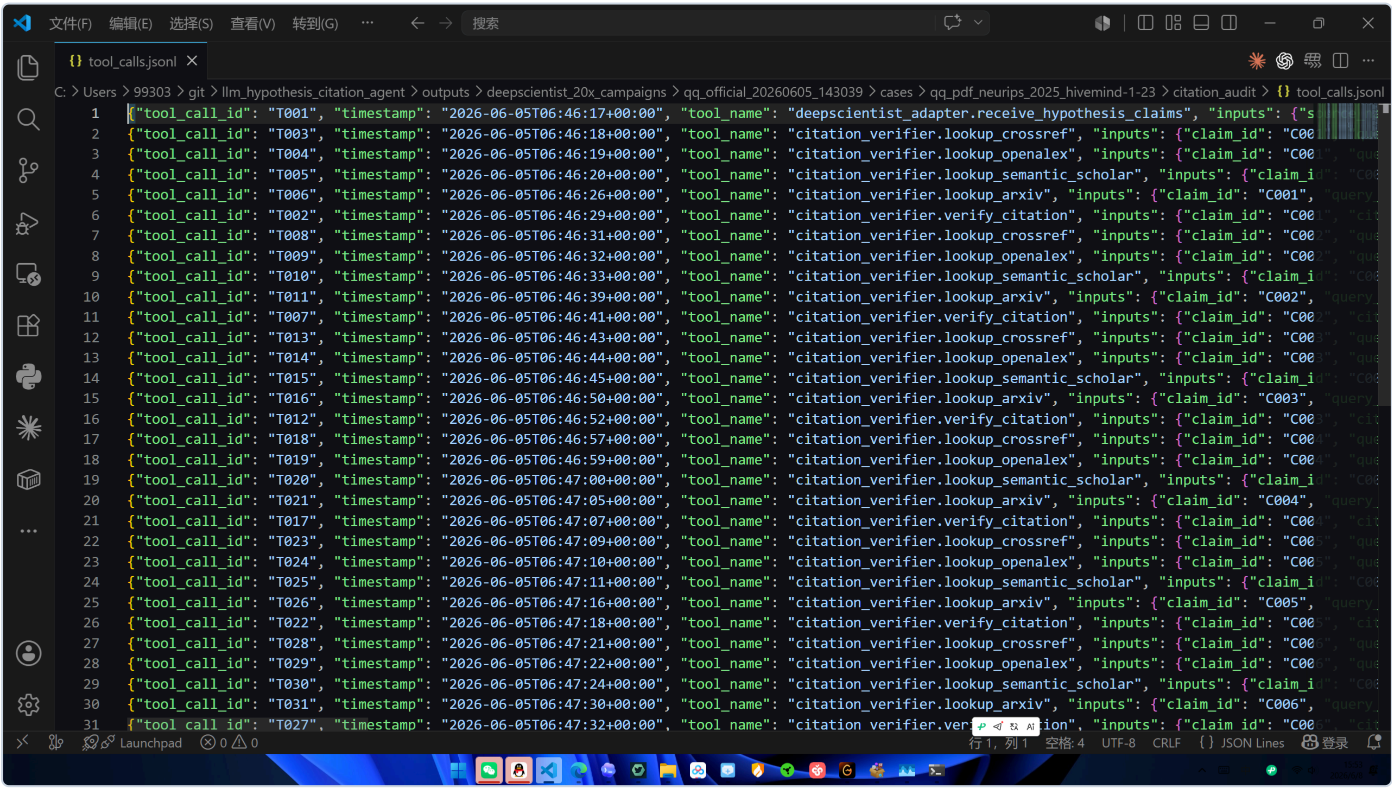

Figure 2. tool_calls.jsonl in VS Code: Crossref, OpenAlex, Semantic Scholar, arXiv, and verifier calls are logged.

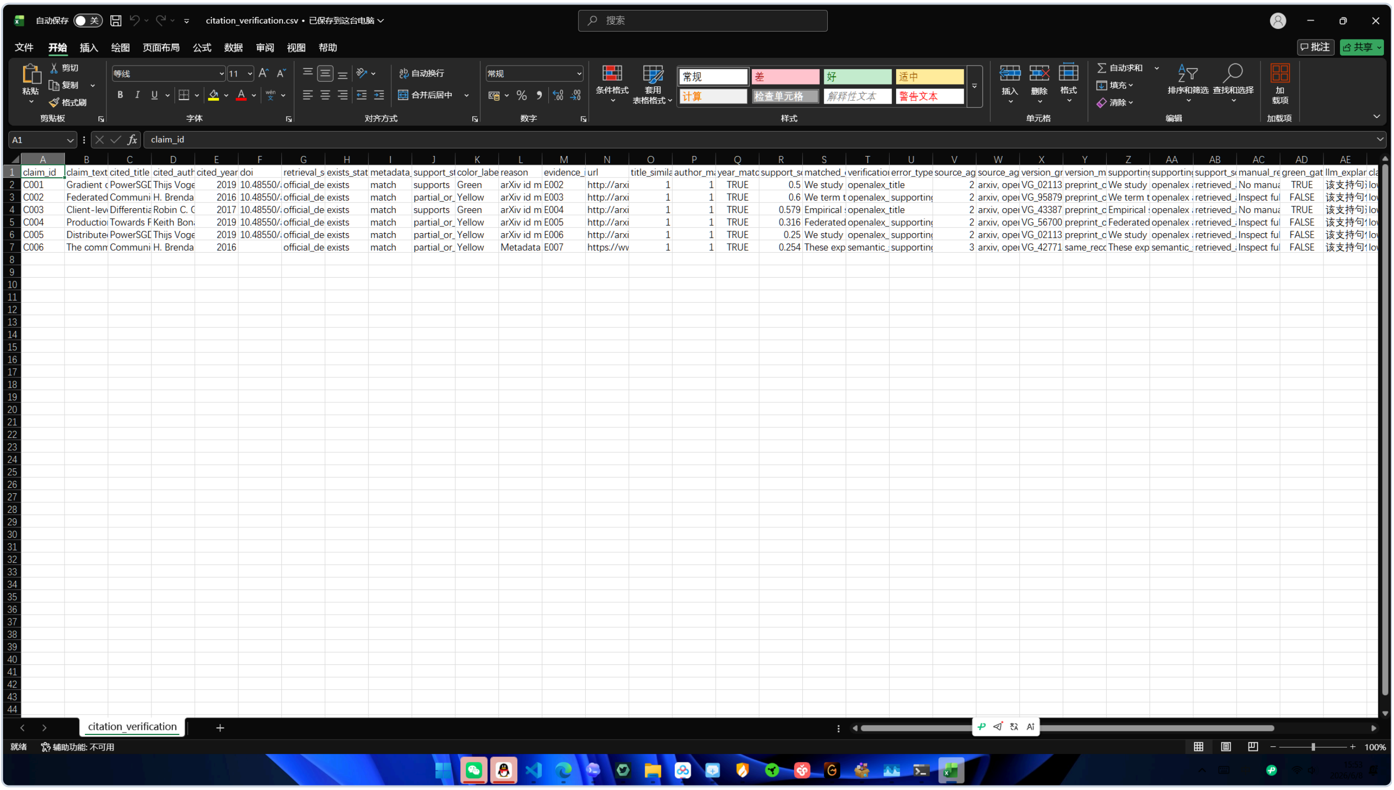

Figure 3. citation_verification.csv in Excel: claim labels, reasons, evidence IDs, and metadata fields are visible.

3. Five Test Samples: Inputs and Output Directories

The five cases below are selected from real local runs. CheXNet input has been normalized into the repository input folder for submission consistency.

Case	Input PDF	Output run directory	G/Y/R	Status
Success case: CLIP / visual-language transfer qq_pdf_ai_clip_transfer	`.../llm_hypothesis_citation_agent/inputs/deepscientist_20_cases/papers/ai_clip_transfer.pdf`	`.../outputs/deepscientist_20x_campaigns/qq_official_20260605_011710/cases/qq_pdf_ai_clip_transfer`	5/1/0	success review=3 revise_before_demo
Test case: Education LLM harm learning qq_pdf_education_llm_harm_learning	`.../llm_hypothesis_citation_agent/inputs/deepscientist_20_cases/papers/education_llm_harm_learning.pdf`	`.../outputs/deepscientist_20x_campaigns/qq_official_20260605_022254/cases/qq_pdf_education_llm_harm_learning`	5/1/0	success review=3 revise_before_demo
Test case: CheXNet medical AI qq_pdf_chexnet_medical_ai	`.../llm_hypothesis_citation_agent/inputs/deepscientist_20_cases/papers/chexnet_medical_ai.pdf`	`.../outputs/deepscientist_20x_campaigns/qq_official_20260605_130909/cases/qq_pdf_chexnet_medical_ai`	1/5/0	success review=2 revise_before_demo
Boundary case: FinRL finance qq_pdf_finance_finrl	`.../llm_hypothesis_citation_agent/inputs/deepscientist_20_cases/papers/finance_finrl.pdf`	`.../outputs/deepscientist_20x_campaigns/qq_official_20260605_040224/cases/qq_pdf_finance_finrl`	0/5/0	success review=2 revise_before_demo
Boundary case: GW150914 gravitational waves qq_pdf_physics_gravitational_waves	`.../llm_hypothesis_citation_agent/inputs/deepscientist_20_cases/papers/physics_gravitational_waves.pdf`	`.../outputs/deepscientist_20x_campaigns/qq_official_20260605_035133/cases/qq_pdf_physics_gravitational_waves`	3/2/1	success review=2 reject_or_boundary_case

Each output directory contains citation audit artifacts, process logs, multi-reviewer results, memory updates, and a final case report.

4. Five Test Samples: Detailed Output Artifact Counts

This table makes the outputs explicit. It shows how many rows or log entries were generated by each part of the pipeline for the selected cases.

Case	Citation CSV rows	Tool calls	Evidence items	Evidence chain rows	Reports present	Key artifact paths
qq_pdf_ai_clip_transfer	6	33	8	6	final_case_report: yes; multi_review_report: yes	`.../cases/qq_pdf_ai_clip_transfer/citation_audit/citation_verification.csv; .../cases/qq_pdf_ai_clip_transfer/citation_audit/tool_calls.jsonl; .../cases/qq_pdf_ai_clip_transfer/citation_audit/evidence_chain.csv; .../qq_official_20260605_011710/cases/qq_pdf_ai_clip_transfer/final_case_report.md`
qq_pdf_education_llm_harm_learning	6	33	8	6	final_case_report: yes; multi_review_report: yes	`.../cases/qq_pdf_education_llm_harm_learning/citation_audit/citation_verification.csv; .../cases/qq_pdf_education_llm_harm_learning/citation_audit/tool_calls.jsonl; .../cases/qq_pdf_education_llm_harm_learning/citation_audit/evidence_chain.csv; .../qq_official_20260605_022254/cases/qq_pdf_education_llm_harm_learning/final_case_report.md`

qq_pdf_chexnet_medical_ai	6	33	8	6	final_case_report: yes; multi_review_report: yes	.../cases/qq_pdf_chexnet_medical_ai/citation_audit/citation_verification.csv; .../cases/qq_pdf_chexnet_medical_ai/citation_audit/tool_calls.jsonl; .../cases/qq_pdf_chexnet_medical_ai/citation_audit/evidence_chain.csv; .../qq_official_20260605_130909/cases/qq_pdf_chexnet_medical_ai/final_case_report.md
qq_pdf_finance_finrl	5	28	7	5	final_case_report: yes; multi_review_report: yes	.../cases/qq_pdf_finance_finrl/citation_audit/citation_verification.csv; .../cases/qq_pdf_finance_finrl/citation_audit/tool_calls.jsonl; .../cases/qq_pdf_finance_finrl/citation_audit/evidence_chain.csv; .../qq_official_20260605_040224/cases/qq_pdf_finance_finrl/final_case_report.md
qq_pdf_physics_gravitational_waves	6	33	8	6	final_case_report: yes; multi_review_report: yes	.../cases/qq_pdf_physics_gravitational_waves/citation_audit/citation_verification.csv; .../cases/qq_pdf_physics_gravitational_waves/citation_audit/tool_calls.jsonl; .../cases/qq_pdf_physics_gravitational_waves/citation_audit/evidence_chain.csv; .../qq_official_20260605_035133/cases/qq_pdf_physics_gravitational_waves/final_case_report.md

5. Five Test Samples: Representative Output Rows

Case	Claim / Cited work	Label	Reason / Evidence
qq_pdf_ai_clip_transfer	Prior work reports that State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. **Cited:** Learning Transferable Visual Models From Natural Language Supervision	**Green**	arXiv id matches public record (2103.00020). A concrete supporting sentence was found (overlap=1.00, similarity=0.90; shared=categories, computer, fixed, object, predetermined, predict, set... **Evidence:** State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. **error_type:** none; **evidence_id:** E002
qq_pdf_education_llm_harm_learning	Prior work reports that This study explores the integration of large language models (LLMs) into educational environments, emphasizing enhanced accessibil... **Cited:** Integrating Large Language Models into Accessible and Inclusive Education: Access Democratization and Individualized Le...	**Green**	Metadata matches public record (title similarity 1.00). A concrete supporting sentence was found (overlap=1.00, similarity=0.94; shared=accessibility, educational, emphasizing, enhanced, en... **Evidence:** This study explores the integration of large language models (LLMs) into educational environments, emphasizing enhanced accessibility, inclusivity, and individ... **error_type:** none; **evidence_id:** E002
qq_pdf_chexnet_medical_ai	Vision Transformers (ViTs) applied directly to sequences of image patches achieve excellent results on image classification benchmarks, matching or exceed... **Cited:** An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	**Green**	Metadata matches public record (title similarity 1.00). A concrete supporting sentence was found (overlap=0.52, similarity=0.16; shared=benchmarks, computational, convolutional, excellent, ... **Evidence:** When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision... **error_type:** none; **evidence_id:** E006
qq_pdf_finance_finrl	Limit Order Book data contains predictive signals for short-term price movements that are not captured by daily OHLCV features. Tsantekidis et al. (2018) ... **Cited:** Using Deep Learning for price prediction by exploiting stationary limit order book features	**Yellow**	Year mismatch: cited 2018, public record 2020. The title/authors/DOI point to the same public record, so this requires manual metadata review. A concrete supporting sentence was found (over... **Evidence:** In this work a new method to construct stationary features is proposed, tested on predicting mid price movements of the Limit Order Book. **error_type:** year_mismatch_requires_review; **evidence_id:** E002
qq_pdf_physics_gravitational_waves	Tests of general relativity with GW150914 placed empirical bounds on high-order post-Newtonian coefficients and constrained the graviton Compton wavelengt... **Cited:** Tests of general relativity with GW150914	**Red**	Author metadata does not overlap with the public record. The evidence is related, but it does not cover critical claim term(s): constrained. overlap=0.38; similarity=0.34; shared=bound, com... **Evidence:** We constrain the graviton Compton wavelength, assuming that gravitons are dispersed in vacuum in the same way as particles with mass, obtaining a 90%-confidenc... **error_type:** author_mismatch; **evidence_id:** E003

6. Success Case: CLIP / Visual-Language Transfer

This is a success case because the system produced multiple Green labels and at least one claim has a concrete supporting sentence, not merely a found title.

Run directory	.../llm_hypothesis_citation_agent/outputs/deepscientist_20x_campaigns/qq_official_20260605_011710/cases/qq_pdf_ai_clip_transfer
Audit result	Green=5, Yellow=1, Red=0
Representative claim	Prior work reports that State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories.
Cited work	Learning Transferable Visual Models From Natural Language Supervision
Supporting sentence	State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories.
Why Green	arXiv id matches public record (2103.00020). A concrete supporting sentence was found (overlap=1.00, similarity=0.90; shared=categories, computer, fixed, object, predetermined, predict, set, state-of-the-art).

A Green label requires existence, metadata match, support_status=supports, and a concrete supporting sentence.

7. Additional Cross-Domain Samples: Education and Medical AI

These cases show that the workflow is not limited to one domain. The inputs include education/LLM and medical AI papers, and both produce auditable claims, citation labels, and evidence IDs.

Case	Domain	G/Y/R	Representative behavior
qq_pdf_education_llm_harm_learning	Education / LLM	5/1/0	Several claims have concrete supporting sentences, so Green is possible; weak or uncertain support remains Yellow.
qq_pdf_chexnet_medical_ai	Medical AI	1/5/0	Many citations are only partially supported, so the verifier conservatively keeps them Yellow instead of forcing Green.

This supports the presentation point: the system prioritizes auditability and conservative verification over optimistic labeling.

8. Boundary Case 1: FinRL All-Yellow Result

FinRL is a boundary case: the workflow completes, papers are found, but no claim is safe enough for Green. The system asks for manual review instead of fabricating certainty.

Run directory	.../llm_hypothesis_citation_agent/outputs/deepscientist_20x_campaigns/qq_official_20260605_040224/cases/qq_pdf_finance_finrl
Audit result	Green=0, Yellow=5, Red=0
Representative claim	Limit Order Book data contains predictive signals for short-term price movements that are not captured by daily OHLCV features. Tsantekidis et al. (2018) demonstrated that a combined CNN-LSTM model trained on stationary LOB features can predict mid-price movements, establishing the informativeness of LOB microstructure for trad...
Cited work	Using Deep Learning for price prediction by exploiting stationary limit order book features
Risk / error type	year_mismatch_requires_review
Verifier reason	Year mismatch: cited 2018, public record 2020. The title/authors/DOI point to the same public record, so this requires manual metadata review. A concrete supporting sentence was found (overlap=0.50, similarity=0.34; shared=book, features, limit, movements, order, price, stationary).

Why this is a boundary case: the verifier can retrieve and inspect sources, but support is not strong enough to safely promote any claim to Green.

9. Boundary Case 2: GW150914 Contains a Red Label

GW150914 shows that Red is not limited to fake DOI. Even if a cited paper exists, metadata conflict or incomplete claim support can invalidate the citation-backed claim.

Run directory	.../llm_hypothesis_citation_agent/outputs/deepscientist_20x_campaigns/qq_official_20260605_035133/cases/qq_pdf_physics_gravitational_waves
Audit result	Green=3, Yellow=2, Red=1
Representative claim	Tests of general relativity with GW150914 placed empirical bounds on high-order post-Newtonian coefficients and constrained the graviton Compton wavelength to a 90% confidence lower bound of 10^13 km, demonstrating that gravitational wave observations can probe strong-field gravity in regimes inaccessible to Solar System tests.
Cited work	Tests of general relativity with GW150914
Red error type	author_mismatch
Supporting sentence	We constrain the graviton Compton wavelength, assuming that gravitons are dispersed in vacuum in the same way as particles with mass, obtaining a 90%-confidence lower bound of 10^{13} km.
Verifier reason	Author metadata does not overlap with the public record. The evidence is related, but it does not cover critical claim term(s): constrained. overlap=0.38; similarity=0.34; shared=bound, compton, confidence, graviton, lower, wavelength.

Why this is a failure/boundary case: the cited paper is real, but metadata and support are not reliable enough, so the claim is marked as a risk item.

10. Where to Take More Backend Screenshots

If the existing backend screenshots are considered insufficient, take additional screenshots from this concrete case directory. These files show the complete pipeline from DeepScientist status, command logs, citation audit, multi-reviewer, memory update, and final report.

Screenshot target	Path	Why it matters
Full case folder tree	`C:\Users\99303\git\llm_hypothesis_citation_agent\outputs\deepscientist_20x_campaigns\qq_official_20260605_130909\cases\qq_pdf_chexnet_medical_ai`	Shows deepscientist / citation_audit / logs / multi_review / memory outputs.
Command and stdout/stderr logs	`C:\Users\99303\git\llm_hypothesis_citation_agent\outputs\deepscientist_20x_campaigns\qq_official_20260605_130909\cases\qq_pdf_chexnet_medical_ai\logs`	Open ds_status_stdout.txt, citation_audit_stdout.txt, multi_reviewer_stdout.txt, memory_update_stdout.txt.
Tool-use proof	`C:\Users\99303\git\llm_hypothesis_citation_agent\outputs\deepscientist_20x_campaigns\qq_official_20260605_130909\cases\qq_pdf_chexnet_medical_ai\citation_audit\tool_calls.jsonl`	Best screenshot for proving real API/tool calls.
Main audit table	`C:\Users\99303\git\llm_hypothesis_citation_agent\outputs\deepscientist_20x_campaigns\qq_official_20260605_130909\cases\qq_pdf_chexnet_medical_ai\citation_audit\citation_verification.csv`	Best screenshot for proving Green/Yellow/Red labels and reasons.
Evidence chain	`C:\Users\99303\git\llm_hypothesis_citation_agent\outputs\deepscientist_20x_campaigns\qq_official_20260605_130909\cases\qq_pdf_chexnet_medical_ai\citation_audit\evidence_chain.csv`	Shows claim_id to evidence_id and tool_call_id mapping.
Reviewer result	`C:\Users\99303\git\llm_hypothesis_citation_agent\outputs\deepscientist_20x_campaigns\qq_official_20260605_130909\cases\qq_pdf_chexnet_medical_ai\multi_review\multi_review_report.md`	Shows multi-reviewer score and risk comments.
Final report	`C:\Users\99303\git\llm_hypothesis_citation_agent\outputs\deepscientist_20x_campaigns\qq_official_20260605_130909\cases\qq_pdf_chexnet_medical_ai\final_case_report.md`	Shows final case status, G/Y/R counts, missing artifacts, failure reason if any.

Recommended screenshot set: one Explorer screenshot of the full case directory tree, one VS Code screenshot of tool_calls.jsonl, one Excel screenshot of citation_verification.csv, and one VS Code screenshot of final_case_report.md.

11. Reproducible Artifact Package

The following files are the main evidence artifacts for every case. They allow the instructor to inspect the tool calls, retrieval records, labels, and final explanation directly.

File	Purpose
`citation_audit/citation_verification.csv`	Main table: cited work, exists_status, metadata_match_status, support_status, Green/Yellow/Red, reason, and evidence_id.
`citation_audit/tool_calls.jsonl`	Tool-call log for Crossref, OpenAlex, Semantic Scholar, arXiv, and verifier logic. This proves the workflow is not prompt-only.
`citation_audit/evidence_items.jsonl`	Evidence objects, including source text, retrieval result, and evidence ID.
`citation_audit/evidence_chain.csv`	Claim-to-evidence mapping with claim_id, evidence_id, tool_call_id, and support category.
`multi_review/multi_review_report.md`	Multi-reviewer assessment of idea quality and citation risk. It cannot override deterministic R/Y/G labels.
`final_case_report.md`	Per-case final status, input PDF, output directory, G/Y/R counts, and failure/boundary reason.

Integrity note: This evidence PDF excludes the /related work citation audit module and only covers the final submitted DeepScientist hypothesis citation audit workflow. Screenshots, CSV files, logs, and case reports come from real local run directories. No DOI, retrieval result, or QQ record is fabricated.